

Priyam Srivastava

Quantum Algorithms, Sensing and Network Control

Ph.D. Student, University of Pittsburgh

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Summary

Physics-trained Ph.D. student developing **variational quantum algorithms** for quantum sensing and **classical reinforcement-learning methods** for quantum-network control. Research combines spin-system simulation, numerical optimization, and analytical performance models to study probe and readout design under interaction and control constraints, and entanglement distribution under finite memory coherence. Earlier research spans measurement-based quantum computation, surface-code encoding, and ultrafast coherent spectroscopy.

Education

Ph.D. in Information Sciences (Telecom)

Jan 2024 – Present; expected 2028

University of Pittsburgh, Department of Informatics and Networked Systems, School of Computing and Information

GPA: 3.846/4.0

- Advisor: Dr. Kaushik P. Seshadreesan.
- Research areas: variational quantum algorithms, quantum sensing and metrology, reinforcement learning, and quantum networking.

B.S.–M.S. Dual Degree in Physics (Integrated)

Jul 2017 – Jun 2022

Indian Institute of Science Education and Research, Bhopal, India

CGPA: 8.10/10.0

- M.S. thesis: *2-Dimensional Coherent Spectroscopy with Chirped and Unchirped Pulses*.
- Advisor: Dr. Rohan Singh, Ultrafast Coherent Spectroscopy Laboratory.

Publications and Manuscripts

Author order follows the corresponding work; my name is in bold. * denotes equal contribution. Links marked “earlier version” refer to public preprints preceding the revised manuscripts listed here.

Conference papers and accepted manuscripts

1. **P. Srivastava**, X. Jin, J. Liu, G. Dutt, T. Purdy, K. Kim, K. P. Seshadreesan. “Variational Joint Magnetometry and Gradiometry on Dipolar Spin Chains.” *Accepted, IEEE Quantum Week (QCE) 2026, QTEM track; to appear.* [arXiv:2605.03906](#).
2. **P. Srivastava**, A. R. Sabavat, S. Jain, *et al.* (incl. K. P. Seshadreesan). “Sequential vs. Synchronized Entanglement Swapping: QKD Secret-Key Rate under Optimized Link-Layer Control.” *Accepted, ACM PQQS 2026; to appear.* Earlier version: [arXiv:2605.04047](#).
3. **P. Srivastava**, V. Kumar, G. Dutt, K. P. Seshadreesan. “Variational Quantum Sensing for Structured Linear Estimation Function.” *IEEE Quantum Week (QCE) 2025, poster track.* [IEEE Xplore: 11250086](#).

Journal submissions

4. X. Jin^{*}, **P. Srivastava**^{*}, R. Wang, *et al.* (incl. K. P. Seshadreesan, J. Liu). “Four Generations of Quantum Biomedical Sensors.” *Submitted to Nature Biomedical Engineering*, 2026. [arXiv:2603.29944](#).
5. M. Chen, Y. Gan, X. Jin, *et al.* (incl. **P. Srivastava**, K. P. Seshadreesan, J. Liu). “When AI Meets Quantum Information: A Comprehensive Review.” *Submitted to Progress in Quantum Electronics*, 2026. [arXiv:2607.00365](#).

Preprints and manuscripts in preparation

6. **P. Srivastava**, V. Kumar, G. Dutt, K. P. Seshadreesan. “Structured Phase Estimation with Dipolar-Interacting Systems: Sequential and Joint Variational Probe-Readout Optimization.” *Revised manuscript in preparation for submission; extends the QCE 2025 work*. Earlier version: [arXiv:2507.22043](#).
7. **P. Srivastava**, V. Katyal, A. Raina. “Encoder Circuit for a Surface Code using a Measurement-Based Quantum Computing Model.” *Preprint*, 2023. [arXiv:2306.10267](#).

Fellowship and Funded Collaborations

- **PQI Graduate Fellowship (2026–2027)** — Pittsburgh Quantum Institute, awarded through the School of Computing and Information; stipend and tuition scholarship for Fall 2026 and Spring 2027.
- **PQI Community Collaboration Award project (2026)** — “Dynamic Programming & Reinforcement Learning for Quantum Network Control (DyReQNet-Ctrl).” Significant contributor. PI: Dr. K. P. Seshadreesan; co-PIs: Profs. P. Krishnamurthy, A. Scheller-Wolf, and S. Tayur.
- **PQI Community Collaboration Award project (2024)** — “NV-Center-Based Quantum Sensing for Biomedical Ultrasound Imaging.” Contributor. PI: Prof. Junyu Liu; co-PIs: Profs. K. P. Seshadreesan, G. Dutt, and K. Kang.

Research Experience

Graduate Student Researcher

Jan 2024 – Present

Quantum Information and Networking Group, University of Pittsburgh

Advisor: Dr. Kaushik P. Seshadreesan

Variational probe and readout optimization

- Developed PennyLane/JAX simulations and classical CMA-ES optimization workflows for 2–8 dipolar-interacting qubits; compared sequential and joint probe-readout training under matched deployed circuit resources [6., 3.].
- Joint optimization improved classical Fisher information (CFI) by 14.9% for uniform encoding at 8 spins and 3 layers.
- Used quantum Fisher information (QFI), CFI, fresh readout optimization, common-Ramsey evaluation, and direct-QFI searches to distinguish probe-selection losses from readout losses.

Joint magnetometry and gradiometry

- Formulated joint magnetic-field and gradient estimation on dipolar spin chains using a Fisher-information determinant objective; analyzed the rank-deficient QFI of GHZ probes [1.].
- Reached 92% of the best-found QFI determinant benchmark with 5 qubits and 3 circuit layers, achieving a 4.2-fold advantage over the standard-quantum-limit determinant.

Reinforcement learning for quantum-network control

- Built a PyTorch/Gymnasium simulator and trained classical REINFORCE link controllers across 24 configurations; compared sequential and synchronized entanglement swapping using identical policies at each operating point [2].
- Identified a coherence-dependent crossover in a four-hop chain: sequential swapping achieved a 19.1% secret-key-rate advantage over synchronized swapping in a tested 5 km-per-link configuration.
- Derived a fidelity-based onset condition for sequential swapping and analyzed how bottleneck position affects delivered rates and the spatial distribution of memory occupancy.
- Identified and addressed an over-purification trap that stalled reinforcement-learning training below the required fidelity threshold.

Collaborative review contributions

- Co-developed a four-generation framework for quantum biomedical sensors; equal-contribution coauthor [4].
- Authored the AI for quantum networking section of *When AI Meets Quantum Information: A Comprehensive Review* [5].

Research Project — MBQC and Surface Codes

Jun 2022 – Jun 2023

IISER Bhopal Advisor: Dr. Ankur Raina

- Developed a surface-code encoder using the measurement-based quantum computing model; performed Qiskit simulations on AWS Braket [7].

M.S. Thesis — Ultrafast Coherent Spectroscopy

Aug 2021 – May 2022

IISER Bhopal Advisor: Dr. Rohan Singh

- Numerically solved optical Bloch equations in Python to study chirped excitation of two- and three-level systems for two-dimensional coherent spectroscopy.
- Studied the theory of two-dimensional coherent spectroscopy and gained bench-side experience with an ultrafast spectroscopy setup.

Summer Research Internship — Entanglement Entropy

May 2019 – Jul 2019

IIT Allahabad Advisor: Dr. Srijit Bhattacharjee

- Studied entanglement entropy in coupled harmonic-oscillator systems.

Talks and Presentations

- **Technical oral presentation**, “Variational Joint Magnetometry and Gradiometry on Dipolar Spin Chains.” IEEE International Conference on Quantum Computing and Engineering (QCE 2026), QTEM track, Toronto, Canada, September 18, 2026.
- **Contributed talk**, “Variational Probe and Measurement Optimization for Structured Phase Estimation.” Workshop on Frontiers in Quantum Information & Technologies (FQIT 2026), Indian Institute of Information Technology Allahabad, March 26–29, 2026.
- **Poster**, “Variational Quantum Sensing for Structured Linear Estimation Function.” IEEE International Conference on Quantum Computing and Engineering (QCE 2025).

Teaching

- **Recitation Instructor and Grader**, CS 1613 *Modern Quantum Computing*, University of Pittsburgh, Spring 2026.
- **Laboratory Teaching Assistant**, CS 0011 *Introduction to Computing for Scientists*, University of Pittsburgh, Spring 2026.
- **Teaching Assistant**, CS 1900 *Internship* and CS 1980 *Team Project*, University of Pittsburgh, Fall 2025 – Spring 2026.

Service

- **Reviewer**, *npj Quantum Information* (Nature Portfolio), 2025.
- **Student Volunteer**, IEEE International Conference on Quantum Computing and Engineering (QCE), 2025 and 2026.

Schools and Professional Development

- **QNumerics 2026**, Numerical Methods in Quantum Information Science, UMass Amherst, June 2026.
- **STAQ Summer School on Quantum Information Theory and Computation**, Duke University, June 2024.

Certifications

- **IBM Quantum Learning**, *Variational Algorithm Design* and *Basics of Quantum Information* badges, 2024–2025.

Technical Skills

Programming and scientific computing: Python, NumPy, SciPy, JAX, PyTorch, Gymnasium; C, MATLAB, Mathematica.

Quantum SDKs: PennyLane, Qiskit, AWS Braket.

Optimization and learning: CMA-ES, gradient-free optimization, REINFORCE, policy gradients.

Research methods: Variational quantum algorithms, parameterized quantum circuits, quantum simulation, classical and quantum Fisher information, multiparameter estimation, quantum-network simulation, stabilizer formalism, measurement-based quantum computation.

Computing infrastructure: Pitt CRC HPC workflows, SLURM array jobs, Git, Linux/bash.

Physical systems and research contexts: Dipolar spin systems, NV centers in diamond, ultrafast and nonlinear optics.

Languages: English (fluent), Hindi (native).